

# A New Lower Bound for the Supremum of Autoconvolutions

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## Abstract

We establish a lower bound on the autoconvolution constant  $C_{1a} = \inf_f \|f * f\|_{L^\infty} / (\int f)^2$ , taken over nonnegative  $f \in L^1(\mathbb{R})$  supported on  $(-1/4, 1/4)$ :

$$C_{1a} \geq 1292/1000 = 1.292.$$

This improves on the previously announced bound of 1.2802 due to Cloninger and Steinerberger [CS17] and on the rigorous analytic bound of 1.27481 established by Matolcsi and Vinuesa [MV10]. The argument extends the Matolcsi–Vinuesa master inequality: the single arcsine kernel is replaced by a convex combination of three arcsine kernels, and the trigonometric multiplier  $G$  is re-optimized as a 200-cosine expansion.

The five resulting functionals are evaluated rigorously in `flint.arb` interval arithmetic at 256-bit precision. The no- $z_1$  quadratic master inequality, applied to the rounded rational anchors, yields strict failure at  $M = 1292/1000$  with margin  $\tau - \Phi(M) \geq 9.6 \times 10^{-5}$ . The analytic reduction is formalized in Lean 4: the headline theorem in `lean/Sidon/MultiScale.lean` reaches a single user axiom (the three-scale MV master inequality, stated at the rational slacks); the quadratic inversion and the slack-soundness statements are Lean theorems (see Section 5).

## 1. Introduction

### 1.1. The Constant $C_{1a}$

Let  $\mathcal{F} = \{f \in L^1(\mathbb{R}) : f \geq 0, \text{ supp}(f) \subseteq (-1/4, 1/4), \int f > 0\}$ . The *autoconvolution constant* is

$$C_{1a} = \inf_{f \in \mathcal{F}} R(f), \quad R(f) := \frac{\|f * f\|_{L^\infty}}{(\int f)^2}.$$

By homogeneity we may normalize  $\int f = 1$ , in which case  $R(f) = \|f * f\|_{L^\infty} \geq 1$ . Heuristically,  $R(f)$  measures the peak height of the self-convolution  $f * f$  relative to its  $L^1$  mass, so  $C_{1a}$  identifies the flattest autoconvolution achievable in the class  $\mathcal{F}$ . This constant is the continuous analogue of the Sidon-set extremal problem [ET41, MO09]; improving its lower bound tightens known asymptotic results on  $B_2^+[1]$  sets and related Sidon variants.

### 1.2. Known Bounds

Prior lower bounds on  $C_{1a}$  are summarized in Table 1. Matolcsi–Vinuesa [MV10] established the rigorous analytic value 1.27481 in 2010, and Cloninger and Steinerberger [CS17] announced a computational improvement to 1.2802 via a discrete branch-and-prune search. All subsequent progress on the constant has been on the upper-bound side, where a recent AI-assisted search lowered the ceiling to 1.5029 [GGTW26]. The present work raises the lower bound to 1.292 (Theorem 1.1), which we refer to as the *Piterbarg–Bajaj–Vincent Bound* (henceforth, the *PBV bound*), by refining the kernel class entering the Matolcsi–Vinuesa dual framework.

| Bound       | Value          | Reference                              |
|-------------|----------------|----------------------------------------|
| Upper bound | $\leq 1.5029$  | Georgiev et al. [GGTW26] (AlphaEvolve) |
| Upper bound | $\leq 1.50992$ | Matolcsi & Vinuesa [MV10]              |
| Lower bound | $\geq 1.262$   | Martin & O’Bryant [MO09]               |
| Lower bound | $\geq 1.27481$ | Matolcsi & Vinuesa [MV10]              |
| Lower bound | $\geq 1.2802$  | Cloninger & Steinerberger [CS17]       |
| Lower bound | $\geq 1.292$   | <b>This work</b>                       |

**Table 1:** Published bounds on  $C_{1a}$  in the normalization used here.

### 1.3. Strategy and Main Result

The Matolcsi–Vinuesa [MV10] dual framework (Section 2) parametrizes a Bochner-admissible kernel  $K$  and a cosine multiplier  $G$ , producing a quadratic lower bound on  $R(f)$  in five real functionals  $(k_1, K_2, S_1, m_G, a)$ , of which  $(k_1, K_2)$  are extracted from the kernel and  $(S_1, m_G, a)$  from the multiplier (with  $a = (4/u) m_G^2/S_1$  a derived gain functional). MV 2010 takes  $K$  to be the arcsine kernel at half-width  $\delta = 138/1000$  and  $G$  a 119-cosine expansion, achieving  $R(f) \geq 1.27481$ . The present construction (i) replaces the single arcsine by a three-scale convex combination  $K_{\text{ms}}$  at half-widths  $(138, 55, 25)/1000$  with weights  $(85, 10, 5)/100$ ; (ii) re-optimizes  $G$  as a 200-cosine expansion against  $K_{\text{ms}}$ ; (iii) certifies the five functionals in `flint.arb` interval arithmetic; and (iv) solves the no- $z_1$  quadratic master inequality with the certified rationals. The quadratic inequality  $\Phi(M) \geq \tau$  fails strictly at the rational witness  $M = 1292/1000$ , with margin  $9.6 \times 10^{-5}$  from the rational anchors, giving the main result.

**Theorem 1.1** (Main result). *Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be nonnegative with  $\text{supp}(f) \subseteq (-1/4, 1/4)$ ,  $\int f > 0$ , and  $\|f * f\|_{L^\infty} < \infty$ . Then*

$$\frac{\|f * f\|_{L^\infty}}{(\int f)^2} \geq \frac{1292}{1000}.$$

*In particular  $C_{1a} \geq 1292/1000 = 1.292$ .*

The proof is assembled in Section 5; all numerical anchors are reproducible from the accompanying certificate, and a Lean 4 mechanization is summarized at the end of Section 5.

## 2. The Matolcsi–Vinuesa Master Inequality

We summarize the dual framework of [MV10, §3]; proofs of the four ingredient lemmas and of the resulting quadratic are recorded in that reference. Throughout,  $f \in \mathcal{F}$  with  $\int f = 1$ ,  $\widehat{h}(\xi) = \int h(x) e^{-2\pi i \xi x} dx$ , and  $M = R(f) = \|f * f\|_{L^\infty}$ . The framework converts the desired lower bound on  $M$  into a quadratic inequality whose coefficients depend on two auxiliary objects of our choice, a positive-definite kernel  $K$  and a trigonometric multiplier  $G$ , so that any rigorous numerical control over  $(K, G)$  translates into a rigorous lower bound on  $M$ .

### 2.1. Admissible Kernels and Multipliers

**Definition 2.1** (Bochner-admissible kernel). Fix  $\delta \in (0, 1/4]$  and set  $u = 1/2 + \delta$ . A function  $K : \mathbb{R} \rightarrow \mathbb{R}$  is *Bochner-admissible at scale  $\delta$*  if

- (K1)  $K \geq 0$  a.e. on  $\mathbb{R}$ ;
- (K2)  $K(-x) = K(x)$  for every  $x \in \mathbb{R}$ ;

(K3)  $\text{supp}(K) \subseteq [-\delta, \delta]$  and  $\int K = 1$ ;

(K4) the periodic Fourier coefficients  $\widetilde{K}(j) := \widehat{K}(j/u)$  satisfy  $\widetilde{K}(j) \geq 0$  for every  $j \in \mathbb{Z}$ .

We write  $K_2 := \|K\|_{L^2(\mathbb{R})}^2$ .

**Definition 2.2** (Admissible cosine multiplier). Fix  $u$  and  $N \in \mathbb{N}$ . A trigonometric polynomial  $G(x) = \sum_{j=1}^N a_j \cos(2\pi jx/u)$  with  $a_j \in \mathbb{R}$  is *admissible* if there is  $m_G > 0$  with  $G(x) \geq m_G$  for every  $x \in [0, 1/4]$ . Define

$$S_1 = \sum_{j=1}^N \frac{a_j^2}{\widetilde{K}(j)}, \quad a = \frac{4}{u} \cdot \frac{m_G^2}{S_1}. \quad (1)$$

## 2.2. The Master Inequality and Its Inversion

**Theorem 2.3** (Master inequality,  $z_1$ -free form). Let  $K$  be Bochner-admissible at scale  $\delta$ ,  $u = 1/2 + \delta$ , and  $G$  admissible of degree  $N$  with constant  $m_G$ , defining  $S_1$  and  $a$  by (1). Then for every  $f \in \mathcal{F}$  with  $\int f > 0$ ,

$$M + 1 + \sqrt{(M-1)(K_2-1)} \geq \frac{2}{u} + a. \quad (2)$$

*Proof.* Matolcsi–Vinuesa [MV10, Eq. (7), Lemma 3.1] prove the sharper form

$$M + 1 + 2z_1 k_1 + \sqrt{(M-1-2z_1^2)(K_2-1-2k_1^2)} \geq \frac{2}{u} + a, \quad (3)$$

where  $k_1 = \widetilde{K}(1)$  and  $z_1 = \widehat{f}(1/u)$ . The  $z_1$ -free designation reflects that  $z_1$  is a Fourier coefficient of the (unknown) extremal  $f$ , so it cannot be controlled analytically; the Cauchy–Schwarz step below absorbs  $z_1$  (together with  $k_1$ ) into the kernel-side quantity  $K_2$ , leaving an inequality that depends only on quantities we can certify. Their proof uses the arcsine kernel only through Bochner-admissibility (Definition 2.1); the same derivation therefore applies to any admissible  $K$ , in particular to the convex combination  $K_{\text{ms}}$ . The radicand of (3) is nonnegative by the same Parseval-type bounds as in [MV10, §3]: Bessel’s inequality gives  $2z_1^2 \leq M-1$  for  $f \in \mathcal{F}$  with  $\int f = 1$ , and  $2k_1^2 \leq K_2-1$  holds for any Bochner-admissible  $K$  with  $\int K = 1$ . Inequality (2) then follows from (3) by the Cauchy–Schwarz estimate  $\sqrt{ac} + \sqrt{bd} \leq \sqrt{(a+b)(c+d)}$  applied to the nonnegative quadruple  $(a, b, c, d) = (2z_1^2, M-1-2z_1^2, 2k_1^2, K_2-1-2k_1^2)$ .  $\square$

**Lemma 2.4** (Quadratic inversion). For fixed  $K_2 \geq 1$ , the function

$$\Phi(M) := M + 1 + \sqrt{(M-1)(K_2-1)}$$

is continuous and strictly increasing on  $[1, \infty)$  with  $\Phi(1) = 2$ . For any  $\tau \geq 2$  the equation  $\Phi(M) = \tau$  has a unique solution  $M_* \in [1, \infty)$ , and every  $f \in \mathcal{F}$  with  $\Phi(R(f)) \geq \tau$  satisfies  $R(f) \geq M_*$ . In particular, with  $\tau = 2/u + a$ ,  $C_{1a} \geq M_*$ .

*Proof.* For  $M > 1$ ,  $\Phi'(M) = 1 + \sqrt{(K_2-1)/(M-1)}/2 > 1$ , so  $\Phi$  is strictly increasing on  $(1, \infty)$  and continuous at  $M = 1$  from the right. Existence, uniqueness, and the conclusion follow from the intermediate value theorem and Theorem 2.3.  $\square$

The MV 2010 single-arcsine optimum yields  $M_* \approx 1.2748$ . The remainder of the paper raises  $M_*$  by enlarging the kernel class.

### 3. The Three-Scale Arcsine Kernel

#### 3.1. Single-Scale Arcsine and the Sonine Identity

For  $\delta \in (0, 1/4]$ , let

$$\eta_\delta(x) := \frac{1}{\delta} \cdot \frac{2/\pi}{\sqrt{1 - (2x/\delta)^2}} \mathbf{1}_{|x| < \delta/2}$$

denote the arcsine density on  $(-\delta/2, \delta/2)$  (the pushforward of the arcsine density on  $(-1/2, 1/2)$  under  $u \mapsto \delta u$ ). By the Sonine–Bessel identity [Wat44, §13.46],  $\widehat{\eta}_\delta(\xi) = J_0(\pi\delta\xi)$ . The *arcsine kernel at scale  $\delta$*  is the auto-convolution

$$K_{\text{arc}}(\delta; \cdot) := \eta_\delta * \eta_\delta,$$

which is nonnegative and even, supported on  $[-\delta, \delta]$ , with  $\int K_{\text{arc}}(\delta; \cdot) = (\int \eta_\delta)^2 = 1$ . The convolution theorem then gives

$$\widehat{K_{\text{arc}}(\delta; \cdot)}(\xi) = \widehat{\eta}_\delta(\xi)^2 = J_0(\pi\delta\xi)^2 \geq 0. \quad (4)$$

In particular each  $K_{\text{arc}}(\delta; \cdot)$  is Bochner-admissible at scale  $\delta$ , with periodic coefficients  $\widetilde{K_{\text{arc}}}(\delta; j) = J_0(\pi j \delta / u)^2$  for  $u \geq 2\delta$ .

#### 3.2. The Convex Combination

**Definition 3.1** (Three-scale kernel). Fix

$$(\delta_1, \delta_2, \delta_3) = \left(\frac{138}{1000}, \frac{55}{1000}, \frac{25}{1000}\right), \quad (\lambda_1, \lambda_2, \lambda_3) = \left(\frac{85}{100}, \frac{10}{100}, \frac{5}{100}\right), \quad (5)$$

and set  $u = 1/2 + \delta_1 = 638/1000$ . The *three-scale arcsine kernel* is

$$K_{\text{ms}}(x) := \sum_{i=1}^3 \lambda_i K_{\text{arc}}(\delta_i; x), \quad x \in \mathbb{R}.$$

| $i$ | $\delta_i$ | $\lambda_i$ | $u$      | $N$ |
|-----|------------|-------------|----------|-----|
| 1   | 138/1000   | 85/100      |          |     |
| 2   | 55/1000    | 10/100      | 638/1000 | 200 |
| 3   | 25/1000    | 5/100       |          |     |

**Table 2:** Parameters of the three-scale arcsine kernel and the cosine multiplier. The period  $u$  and the cosine degree  $N$  are shared across the three scales.

#### 3.3. Admissibility

**Theorem 3.2** (Admissibility). *The kernel  $K_{\text{ms}}$  is Bochner-admissible at scale  $\delta_1 = 138/1000 < 1/4$ , with periodic coefficients*

$$\widetilde{K_{\text{ms}}}(j) = \sum_{i=1}^3 \lambda_i J_0\left(\frac{\pi j \delta_i}{u}\right)^2 \geq 0, \quad j \in \mathbb{Z}. \quad (6)$$

*Proof.* We check conditions (K1)–(K4) of Definition 2.1.

- (K1) *Nonnegativity*, (K2) *evenness*. Each  $\lambda_i K_{\text{arc}}(\delta_i; \cdot) \geq 0$  and is even; nonnegative sums of even nonnegative functions are even and nonnegative.
- (K3) *Support and unit mass*.  $\text{supp } K_{\text{arc}}(\delta_i; \cdot) \subseteq [-\delta_i, \delta_i] \subseteq [-\delta_1, \delta_1]$  for  $i \geq 1$ , so  $\text{supp } K_{\text{ms}} \subseteq [-\delta_1, \delta_1]$ ; and  $\int K_{\text{ms}} = \sum_i \lambda_i \int K_{\text{arc}}(\delta_i) = \sum_i \lambda_i = 1$ .
- (K4) *Fourier positivity*. Linearity of the Fourier transform and (4) give

$$\widehat{K_{\text{ms}}}(\xi) = \sum_i \lambda_i J_0(\pi \delta_i \xi)^2 \geq 0,$$

hence (6).  $\square$

The motivation is structural: in (1),  $S_1$  blows up at frequencies  $j$  where the single-scale  $J_0(\pi j \delta / u)^2$  is close to a Bessel zero. Mixing three scales shifts the zeros of each summand, so the sum stays bounded away from zero on  $1 \leq j \leq 200$  (numerically  $\min_j \widehat{K_{\text{ms}}}(j) \geq 2.08 \times 10^{-4}$ ). This drops  $S_1$  from  $\approx 87.4$  (single-scale) to  $\leq 29.841$  (Lemma 4.3), and proportionally raises the gain  $a$ .

## 4. The Rigorous Numerical Certificate

### 4.1. The Five Certified Anchors

For the master inequality applied to  $(K_{\text{ms}}, G)$  we need rigorous bounds on five real functionals. We replace each by a rational *anchor*: a floor when the functional enters the master inequality on the side that weakens the bound under understatement (i.e. with a  $\geq$  sign), a ceiling on the side that weakens it under overstatement ( $\leq$ ), in both cases obtained by outward rounding of the certifier's arb interval enclosure. Anchoring at exact rationals lets the final inversion (Section 5) be carried out in exact rational arithmetic, independent of the floating-point pipeline that produced the enclosures. The cosine multiplier  $G(x) = \sum_{j=1}^N a_j \cos(2\pi j x / u)$  with  $N = 200$  has rational coefficients  $a_j \in \mathbb{Q}$  with denominator  $10^8$ , obtained by rounding the solution of the convex QP that minimizes  $\sum_j a_j^2 / \widehat{K_{\text{ms}}}(j)$  subject to  $G \geq 1$  on a uniform grid of 5001 points in  $[0, 1/4]$  (solved by MOSEK). The five lemmas below are arb-interval computations [Joh17]; all certified outputs are rounded outward to exact rationals. Only four of the five anchors ( $K_2$ ,  $S_1$ ,  $m_G$ ,  $a$ ) enter the  $z_1$ -free form (2) used in Section 5; the kernel-mass moment  $k_1$  is recorded because the certifier internally uses the sharper inequality (3) (which sharpens  $M_{\text{cert}}$  from 1.292 to  $\approx 1.29232$  before rounding to the rational headline target).

**Lemma 4.1** (Kernel-mass moment).  $k_1 := \widehat{K_{\text{ms}}}(1) = \sum_{i=1}^3 \lambda_i J_0(\pi \delta_i)^2 \geq 9212/10000$ .

*Proof.* At 256-bit precision, `arb.bessel_j(0)` evaluated at exact rational  $\pi \delta_i$  returns balls of radius  $< 7 \times 10^{-77}$ ; the certified sum is  $k_1 \geq 0.92124658$ , with slack  $\geq 4.6 \times 10^{-5}$  above 9212/10000.  $\square$

**Lemma 4.2** (Kernel-energy moment).  $K_2 = \|K_{\text{ms}}\|_{L^2(\mathbb{R})}^2 \leq 47897/10000$ .

*Proof.* Split  $K_2 = 2 \int_0^T \widehat{K_{\text{ms}}}(\xi)^2 d\xi + 2 \int_T^\infty \widehat{K_{\text{ms}}}(\xi)^2 d\xi$  with  $T = 10^5$ . The bulk integral is computed by arb adaptive Gauss–Legendre quadrature and certified at 4.78882342... with arb radius below  $10^{-10}$ . For the tail, the standard uniform bound  $|J_0(z)|^2 \leq 2/(\pi z)$  [Wat44, §7.21] holds for  $z \geq 1$ ; since  $\xi \geq T = 10^5$  in the tail forces  $z = \pi \delta_i \xi \geq \pi(25/1000)(10^5) \approx 7854$ , the bound applies, and applied to both Bessel factors gives  $J_0(\pi \delta_i \xi)^2 J_0(\pi \delta_j \xi)^2 \leq 4/(\pi^4 \delta_i \delta_j \xi^2)$ , whence  $\widehat{K_{\text{ms}}}(\xi)^2 = \sum_{i,j} \lambda_i \lambda_j J_0(\pi \delta_i \xi)^2 J_0(\pi \delta_j \xi)^2 \leq 4C^2/(\pi^4 \xi^2)$  with  $C = \sum_i \lambda_i / \delta_i \approx 9.978$ . Integrating  $\int_T^\infty \xi^{-2} d\xi = 1/T$  and doubling for both half-lines,  $2 \int_T^\infty \widehat{K_{\text{ms}}}^2 \leq 8C^2/(\pi^4 T) \leq 8.19 \times 10^{-5}$ . Summing,  $K_2 \in [4.78882342, 4.78890519] \subseteq [0, 47897/10000]$ , slack  $\geq 7.9 \times 10^{-4}$ .  $\square$

**Lemma 4.3** (Multiplier denominator).  $S_1 = \sum_{j=1}^{200} \frac{a_j^2}{\widetilde{K_{\text{ms}}}(j)} \leq 29841/1000$ .

*Proof.* Each  $\widetilde{K_{\text{ms}}}(j)$  is evaluated in arb from (6) with radius  $< 10^{-70}$ ; dividing exact rational squares  $a_j^2$  yields a certified sum  $S_1 \leq 29.840907$ , slack  $\geq 9.3 \times 10^{-5}$ .  $\square$

**Lemma 4.4** (Pointwise lower bound of  $G$ ).  $m_G := \min_{x \in [0, 1/4]} G(x) \geq 998/1000$ .

*Proof.* Partition  $[0, 1/4]$  into  $n = 32768$  closed cells of half-width  $r = 1/(8n) \approx 3.81 \times 10^{-6}$ . On each cell centred at  $c$ , the second-order Taylor expansion of  $G$  yields the rigorous arb enclosure

$$G(\text{cell}) \subseteq G(c) + G'(c) \cdot [-r, r] + G''(\text{cell}) \cdot [-r^2/2, r^2/2],$$

with  $G''$  evaluated as an arb interval over the entire cell from

$$G''(x) = - \sum_{j=1}^{200} a_j (2\pi j/u)^2 \cos(2\pi jx/u).$$

The minimum of the lower endpoints over all  $n$  cells gives the certified bound  $m_G \geq 0.99997987$ , slack  $\geq 1.9 \times 10^{-3}$  above  $998/1000$ .  $\square$

**Lemma 4.5** (Gain functional).  $a = (4/u) \cdot m_G^2/S_1 \geq 20925/100000$ .

*Proof.* Substituting the rational anchors of Lemmas 4.3 and 4.4 into the definition of  $a$  in (1),

$$a \geq \frac{4}{638/1000} \cdot \frac{(998/1000)^2}{29841/1000} = \frac{4000 \cdot 998^2}{638 \cdot 29841 \cdot 10^3} = \frac{3984016000}{19038558000} = 0.20926\dots,$$

hence  $a \geq 20925/100000$ , slack  $\geq 1.0 \times 10^{-5}$ . (The underlying arb interval evaluation returns the tighter enclosure  $a \geq 0.21009214$  when  $m_G$  and  $S_1$  are coupled rather than rounded separately; we use only the looser rational floor in what follows.)  $\square$

| Functional             | Symbol | Direction | Rational bound | Decimal |
|------------------------|--------|-----------|----------------|---------|
| Kernel-mass moment     | $k_1$  | $\geq$    | 9212/10000     | 0.9212  |
| Kernel-energy          | $K_2$  | $\leq$    | 47897/10000    | 4.7897  |
| Multiplier denominator | $S_1$  | $\leq$    | 29841/1000     | 29.841  |
| Multiplier minimum     | $m_G$  | $\geq$    | 998/1000       | 0.998   |
| Gain                   | $a$    | $\geq$    | 20925/100000   | 0.20925 |

**Table 3:** Certified rational anchors for the three-scale kernel  $K_{\text{ms}}$  at parameters (5) with  $u = 638/1000$  and the 200-cosine multiplier  $G$ .

## 4.2. Rescaling Invariance

Although the proof of Lemma 4.4 produces  $m_G \geq 998/1000$ , the master inequality is invariant under  $G \mapsto G/m_G$ : rescaling multiplies both  $S_1$  and  $m_G^2$  by  $1/m_G^2$ , leaving  $a$  unchanged. The certified rational bound  $m_G \geq 998/1000$  is therefore sufficient for the conclusion, and the lemma is not sharpened further.

### 4.3. Reproducibility

All five anchors are produced by the reproducible script and the associated certificate JSON: each entry is the outward-rounded enclosure returned by `flint.arb` at 256-bit precision; full pipeline inputs (the rational  $a_j$ , the bisection schedule, the quadrature tolerance) and the auditor logs are included in the accompanying repository.

## 5. Proof of the Main Theorem

### 5.1. The Strict-Failure Witness

We close the master inequality using the anchors of Table 3. The strategy is to exhibit a rational *witness*  $M_0 = 1292/1000$  for which  $\Phi(M_0) < \tau$  holds strictly; combined with the master inequality  $\Phi(R(f)) \geq \tau$ , this rules out  $R(f) \leq M_0$  by strict monotonicity of  $\Phi$  (Lemma 2.4), forcing  $R(f) > M_0$ . No analytic optimization over  $M$  is needed: the rational  $M_0$  is fixed in advance and the verification is purely arithmetic. Set  $\tau := 2/u + a$  and recall  $\Phi(M) := M + 1 + \sqrt{(M-1)(K_2-1)}$  from Lemma 2.4. With the certified rationals from Table 3,

$$\tau \geq \frac{2}{638/1000} + \frac{20925}{100000} = 3.134796\dots + 0.20925 = 3.344046\dots$$

**Proposition 5.1** (Strict failure at the rational witness). *With  $K_2 \leq 47897/10000$  and  $\tau = 2/u + a \geq 2000/638 + 20925/10^5$ ,*

$$\Phi(1292/1000) \leq 3.34395 < 3.344046\dots \leq \tau, \quad (7)$$

*with margin  $\tau - \Phi(1292/1000) \geq 9.6 \times 10^{-5}$ .*

*Proof.* At  $M = 1292/1000$ , using the certified upper bound  $K_2 \leq 47897/10000$ ,

$$(M-1)(K_2-1) \leq \frac{292}{1000} \cdot \frac{37897}{10000} = \frac{11065924}{10^7} = 1.1065924,$$

and since  $\Phi$  is increasing in  $K_2$ , this upper bound transfers to  $\Phi(M)$ . The rational bound  $(105195/10^5)^2 = 11065988025/10^{10} > 11065924/10^7$  gives  $\sqrt{(M-1)(K_2-1)} \leq 105195/10^5 = 1.05195$ , so

$$\Phi(1292/1000) \leq \frac{1292}{1000} + 1 + \frac{105195}{10^5} = \frac{66879}{20000} = 3.34395.$$

Combining with  $\tau \geq 4267003/1276000 = 3.344046\dots$  yields

$$\tau - \Phi(1292/1000) \geq \frac{4267003}{1276000} - \frac{66879}{20000} = \frac{307}{3190000} \geq 9.6 \times 10^{-5} > 0.$$

The conclusion is independently re-certified by an arb cell-search bisection at 256-bit precision, which returns the sharper enclosure  $M_{\text{cert}} \geq 1.29215$  when the script's tighter (non-rational) interval bounds on  $m_G$  and  $S_1$  are used.  $\square$

### 5.2. Assembly

**Theorem 5.2** (Restatement of Theorem 1.1). *Let  $f \in \mathcal{F}$  with  $\|f * f\|_{L^\infty} < \infty$ . Then  $R(f) \geq 1292/1000$ , and hence  $C_{1a} \geq 1292/1000$ .*

*Proof.* By homogeneity assume  $\int f = 1$ , so  $M := R(f) = \|f * f\|_{L^\infty} \geq 1$ . We assemble in five steps.



- (1) *Admissibility.* By Theorem 3.2,  $K_{\text{ms}}$  is Bochner-admissible at scale  $\delta_1 = 138/1000$ , with period  $u = 638/1000$ .
- (2) *Anchors.* By Lemmas 4.1–4.5, the multiplier  $G$  and the five rational bounds  $k_1 \geq 9212/10000$ ,  $K_2 \leq 47897/10000$ ,  $S_1 \leq 29841/1000$ ,  $m_G \geq 998/1000$ , and  $a \geq 20925/100000$  hold.
- (3) *Master inequality.* Theorem 2.3 applied to  $(K_{\text{ms}}, G)$  yields  $\Phi(M) \geq \tau$ . Since  $\Phi$  is increasing in  $K_2$ , replacing the unknown  $K_2$  by the certified upper bound  $47897/10000$  overestimates  $\Phi$ ; the strict failure  $\Phi(M_0) < \tau$  at the witness  $M_0 = 1292/1000$  therefore implies  $\Phi(R(f)) > \Phi(M_0)$ , which (by Lemma 2.4) forces  $R(f) > M_0$ .
- (4) *Strict failure.* By Proposition 5.1,  $\Phi(1292/1000) < \tau$ .
- (5) *Conclusion.* Strict monotonicity of  $\Phi$  on  $[1, \infty)$  (Lemma 2.4) together with (3) and (4) forces  $M > 1292/1000$ . Taking the infimum over  $f \in \mathcal{F}$  gives  $C_{1a} \geq 1292/1000 = 1.292$ .  $\square$

### 5.3. Lean Formalization

The analytic chain above is mechanized in Lean 4 in the file `lean/Sidon/MultiScale.lean`, which builds cleanly under Mathlib [The20, dMU21] with zero `sorry` tactics and produces the headline theorem. The headline theorem reaches exactly *one* user axiom in its dependency closure: `MV_master_inequality_for_extremiser`, the three-scale extension of [MV10, Lemma 3.1] (whose proof is a verbatim repeat of the single-scale Fourier reduction with each  $J_0(\pi\delta\xi)^2$  replaced by  $\sum_i \lambda_i J_0(\pi\delta_i\xi)^2$ ), stated with the rational slacks `K2UpperQ` = 47897/10000 and `gainLowerQ` = 20925/100000 substituted for  $K_2$  and  $a$  respectively. This substitution is sound because the master inequality is monotone in  $K_2 - 1$  and  $a$ , and the slack rationals are true bounds on the analytic functionals: those bounds (Lemmas 4.1–4.5) are discharged externally by the `flint.arb` certifier in `delsarte_dual/grid_bound_alt_kernel/`.

Six further statements are Lean *theorems*: the algebraic inversion `master_inequality_M_lower` (corresponding to Proposition 5.1, proved by case analysis on  $M \leq 1$  versus  $M > 1$  using `Real.sqrt` monotonicity) and the five slack-soundness theorems `K_two_upper_bound`, `k_one_lower_bound`, `S_one_upper_bound`, `min_G_lower_bound`, `gain_lower_bound`, each a one-line `norm_num` verification that the rational slack is on the correct side of the certifier-reported decimal recorded in Lemmas 4.1–4.5.

An explicit finiteness hypothesis  $\|f * f\|_{L^\infty} < \infty$  appears in the Lean statement because of the `ENNReal.toReal` encoding; this is harmless for the infimum. The support hypothesis is stated as `Function.support f`  $\subseteq (-1/4, 1/4)$ , i.e.  $f$  vanishes pointwise off  $(-1/4, 1/4)$ ; for any  $f \in \mathcal{F}$  in the sense of Section 1.1, its representative obtained by zero-extending off the essential support satisfies the pointwise condition and has the same autoconvolution ratio, so the Lean bound transfers to the  $L^1$ -equivalence-class formulation of  $C_{1a}$  without loss.

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